

# Introduction to Nutanix and AOS

## day-2

### Boot-up of a CVM - continued...

Once the kexec happens and kernel from svmboot.iso is shifted from the kernel in SSD, svmboot.iso's purpose has been served. If this ISO gets accidentally deleted when the CVM is running, there is no impact, and you can simply restore it by either copying it over from another host, or by copying the **svmboot.iso** from **~/data/installer/AOS\_VERSION/goldimages/** directory on any CVM in the cluster.

Once the CVM starts up from the kernel on the SSD, we go through the usual Linux way of doing things. Systemctl is used to examine/monitor and control the state of 'systemd'. So, systemctl manages systemd, which invokes a Nutanix-specific service called **nutanix-rc-local.service**

```
nutanix@NTNX-16SM65230111-A-CVM:10.136.106.126:~$ systemctl l
ist-unit-files --type=service --state=enabled
UNIT FILE                                STATE
auditd.service                          enabled
autovt@.service                         enabled
crond.service                           enabled
getty@.service                           enabled
haveged.service                         enabled
ip6tables.service                       enabled
ipmi.service                            enabled
ipmievdev.service                       enabled
ipset.service                           enabled
iptables.service                       enabled
irqbalance.service                     enabled
mdmonitor.service                       enabled
ntpd.service                            enabled
nutanix-rc-local.service                 enabled
```

|                                   |         |
|-----------------------------------|---------|
| openibd.service                   | enabled |
| rhel-autorelabel-mark.service     | enabled |
| rhel-autorelabel.service          | enabled |
| rhel-configure.service            | enabled |
| rhel-dmesg.service                | enabled |
| rhel-domainname.service           | enabled |
| rhel-import-state.service         | enabled |
| rhel-loadmodules.service          | enabled |
| rhel-readonly.service             | enabled |
| rpcbind.service                   | enabled |
| rsyslog.service                   | enabled |
| smartd.service                    | enabled |
| sshd.service                      | enabled |
| sysstat.service                   | enabled |
| systemd-readahead-collect.service | enabled |
| systemd-readahead-drop.service    | enabled |
| systemd-readahead-replay.service  | enabled |
| vgauthd.service                   | enabled |
| vmtoolsd.service                  | enabled |

This service, when it starts running, executes **/etc/nutanix/rc.local**

```

nutanix@NTNX-16SM65230111-A-CVM:10.136.106.126:~$ systemctl c
at nutanix-rc-local.service
# /usr/lib/systemd/system/nutanix-rc-local.service
[Unit]
Description=Nutanix rc.local compatibility
After=multi-user.target
[Service]
Type=oneshot
ExecStart=/etc/nutanix/rc.local
GuessMainPID=no
KillMode=none

```

This rc.local in turn invokes anything located in the directory **/usr/local/nutanix/start.d/** and logs everything in **/tmp/startd.log**

```
nutanix@NTNX-16SM65230111-A-CVM:10.136.106.126:~$ cat /etc/nutanix/rc.local
#!/bin/sh
#
# This script will be executed *after* all the other init scripts.
# You can put your own initialization stuff in here if you don't
# want to do the full Sys V style init stuff.
set -e
# start rpcbind
service rpcbind restart
if [ -x "/usr/local/nutanix/rc.nutanix" ]; then
/usr/local/nutanix/rc.nutanix >> /tmp/rc.nutanix.log <-- Used Post Upgrades and it runs various post installation scripts and commands; Only present after upgrades
fi
if [ -d "/usr/local/nutanix/start.d" ]; then
for file in /usr/local/nutanix/start.d/*
do
if [ -x "$file" ]; then <-- Executing every file inside start.d
echo $file >> /tmp/startd.log
"$file" >> /tmp/startd.log
echo >> /tmp/startd.log
fi
done
fi
touch /var/lock/subsys/local
```

Within **/usr/local/nutanix/start.d/**, there's another rc.local defined. This is where all the AOS magic begins. It's a big script which checks for multiple things and

proceeds towards starting the services:

```
nutanix@NTNX-16SM65230111-A-CVM:10.136.106.126:~$ ls -ahl /usr/local/nutanix/start.d
total 20K
drwxr-x---.  2 nutanix nutanix 4.0K Mar 30  2022 .
drwxr-x---. 34 nutanix nutanix 4.0K Nov 21 09:56 ..
rwxr-x---.  1 nutanix nutanix 11K Mar 30  2022 rc.local
```

With the context of our discussion, it first start Hades and run other script that need to be executed before Genesis is brought up:

```
# If hades is installed, start it.
echo "Starting Hades"
/usr/local/nutanix/bootstrap/bin/hades start
# Run any scripts that need to be run after hades is up and
# before genesis starts.
if [ -d "/home/nutanix/hades/post_hades" ]; then
for file in /home/nutanix/hades/post_hades/*; do
if [ -x "$file" ]; then
echo Starting $file at date`
"$file"
rc=$?
echo Finished $file rc:${rc} at date`
if [ $rc -ne 0 ]; then
# If we are not able to process the post hades scripts, we
# should abort this script.
echo "Exiting rc.local due to rc=$rc"
exit $rc
fi
fi
done
fi
.
.
```

```
.  
.br/># If genesis is installed, we'll just start that - it'll take  
care of the rest.  
# Otherwise, we'll start the software the old-fashioned way.  
if [ -x /home/nutanix/cluster/bin/genesis ] ;then  
echo "Starting Genesis"  
runcon -u unconfined_u -t unconfined_t -r unconfined_r \\  
su - nutanix -c "/home/nutanix/cluster/bin/genesis start"  
else  
echo "Not starting Nutanix services as it is not installed."  
fi
```

## Genesis

Port: 2100

Genesis is responsible for bootstrapping a Nutanix cluster and managing the cluster after creation. It runs on each node and is responsible for service interactions (start/stop/etc.).

### Responsibilities of Genesis Includes:

- Discover other instances of Genesis
- Starting/stopping Nutanix services
- Upgrading software
- Storage HA
- Cluster Management
- Handles IP configuration tasks
- Time synchronisation over NTP
- Workflows:
  - 1-click upgrades

- Cluster expansion
- Single SSD repair
- Cluster conversion
- Host Boot Disk Repair

## If node is un-configured:

- Genesis and Foundation are the only services running
- Publishes node information so un-configured nodes can be discovered

All genesis activities is logged in: `/home/nutanix/data/logs/genesis.out`

## Service Management with Genesis

1. "`gs`" or "`genesis status`": This will show all the services that are currently running on a CVM along with their PID

```
nutanix@NTNX-16SM65230111-A-CVM:10.136.106.126:~$ genesis
status
2023-05-08 16:15:21.617332: Services running on this node:
acropolis: [11033, 11097, 11098, 11099]
aequitas: [13185, 13317, 13318, 13364]
alert_manager: [10477, 10611, 10612, 10722]
anduril: [13612, 13777, 13778, 13779]
aplos: [13534, 13601, 13602, 13605]
aplos_engine: [13203, 13347, 13348, 13349]
arithmos: [10501, 10712, 10713, 10794]
athena: [7884, 8003, 8004, 8005]
cassandra: [7354, 7390, 7391, 7411, 7703]
catalog: [10560, 10775, 10776, 10777]
cerebro: [10088, 10211, 10212, 10281]
chronos: [10103, 10222, 10223, 10235]
cim_service: [10464, 10571, 10572, 10577, 10578]
cluster_config: [13173, 13274, 13275, 13276]
cluster_health: [13703, 13858, 13955, 13956, 13964, 13977,
14020, 14022, 14024, 14026, 14027, 14040, 14041, 14043, 14
```

051, 14053, 14057, 14112, 14113, 14117, 14155, 14156, 1416  
8, 14169, 14177, 14178, 14179, 14180, 14292, 14293, 14295,  
14296, 14298, 14299, 14300, 14301, 14302, 14303, 14305, 14  
313, 14315, 14329, 14396, 14400]  
curator: [10271, 10327, 10328, 24926]  
delphi: [13550, 13692, 13693, 13694]  
dynamic\_ring\_changer: [7865, 7965, 7966, 8127]  
ergon: [10073, 10149, 10150, 10151]  
flow: [13572, 13736, 13737, 13739]  
foundation: []  
genesis: [10418, 10504, 10529, 10530, 14207, 14208, 14209,  
14212]  
ikat\_control\_plane: [6945, 7017, 7018, 7019]  
ikat\_proxy: [6662, 6710, 6711, 6712]  
insights\_data\_transfer: [9862, 9931, 9932, 9939, 9940, 994  
1, 9942]  
insights\_server: [7880, 8072, 8073, 8149]  
lazan: [952, 1634, 1635, 13541, 13650, 13651, 18887]  
mantle: [7911, 8098, 8099, 8128]  
mercury: [7898, 8125, 8126, 8184]  
minerva\_cvm: [13164, 13234, 13235, 13236, 13473]  
nutanix\_guest\_tools: [11282, 11359, 11360, 11378, 11404]  
pithos: [7870, 8035, 8036, 8059]  
prism: [10455, 10538, 10539, 10624, 10626, 10631, 10663, 1  
0664, 10665]  
scavenger: [13805, 13917, 13918, 13920]  
secure\_file\_sync: [6956, 7104, 7105, 7106]  
snmp\_manager: [11261, 11307, 11308, 11309]  
ssl\_terminator: [6950, 7064, 7065]  
stargate: [9603, 9634, 9635, 9636, 9637]  
sys\_stat\_collector: [6011, 6073, 6074, 6075]  
uhura: [11146, 11197, 11198, 11199]  
xmount: [3792, 3826, 3827, 3872]  
xtrim: [13659, 13825, 13826, 13827]  
zookeeper: [25834, 25848, 27544, 27608, 27609, 27644]

- 
2. `"genesis restart"` : Restarts genesis service
  3. `"genesis stop"` : Stops the genesis service
  4. `"genesis stop <service name>"` : Stops a service
  5. `"genesis stop all"` : To stop all services in a CVM
  6. `"cluster start"` : Starts any services which are down

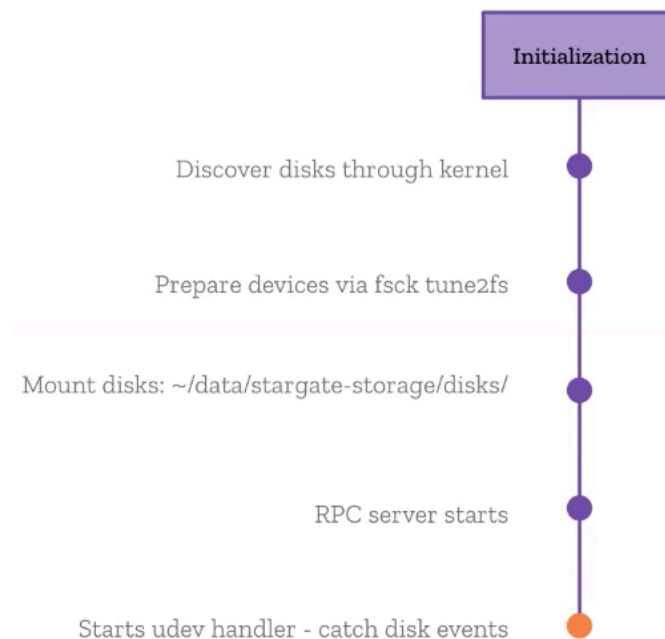
Note: To start a stopped Zookeeper and Scavenger, we need to perform a genesis restart

When a cluster stop command is run, other services will stop however scavenger and zookeeper

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## Hades

Port: 2099



- Hades is the process that manages the state of our disks for the CVMs.



- Hades does things like check disk health and mount/unmount disks on a particular CVM.
- Hades is one of the first processes to start on a CVM.

Hades loggings are done in: `/home/nutanix/data/logs/hades.out`

## Common Hades Commands

Check status and PIDs of Hades: `sudo /usr/local/nutanix/bootstrap/bin/hades status`

Stop hades: `sudo /usr/local/nutanix/bootstrap/bin/hades stop`

Start hades: `sudo /usr/local/nutanix/bootstrap/bin/hades start`

Restart hades: `sudo /usr/local/nutanix/bootstrap/bin/hades restart`

Note: Hades restart will not touch already mounted, partitioned, formatted disks.. It will mount/partition/format any disks which is missing any of these

## Responsibilities

1. Disk removal and addition - graceful, hotplugging
2. Disk offline - if stargate experiences very high latency it updates the zeus\_proto with offline path then it marks it as offline disk, offline paths are monitored by genesis and it creates a RPC to hades, a possible reason could be that hades is node local and genesis is cluster level.
3. Disk diagnostics - if SMART values are okay then hades marks disk good else it marks it bad
4. Tombstone disks
5. DDD - degraded disk detection

## Hades Proto

Hades proto is the disk details stored by hades in json format in

zookeeper: `/appliance/physical/hades/configuration/<svm-id>`

The hades proto can be printed in a human readable format using the following command: `edit-hades -p`

```

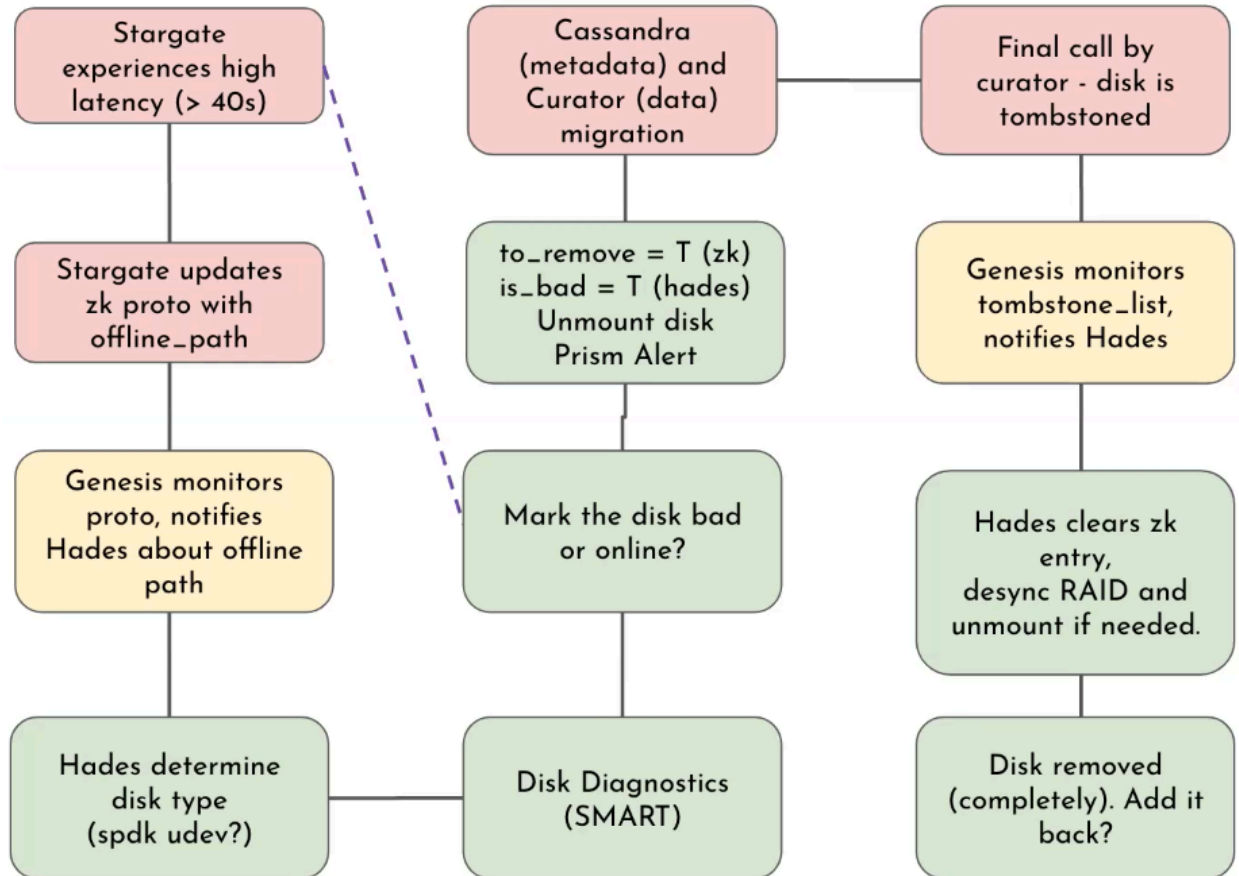
nutanix@NTNX-16SM65230111-A-CVM:10.136.106.126:~$ edit-hades
-p
No cvm-id provided. Using cvm-id:7 of current node
slot_list {
location: 1
disk_present: true
disk {
serial: "BTHC618600FX800NGN"
model: "INTEL SSDSC2BX800G4"
is_mounted: true
current_firmware_version: "0140"
target_firmware_version: "0140"
is_boot_disk: true
vendor_info: "Not Available"
raw_capacity: 800166076416
}
logical_slot_id: 1
}
slot_list {
location: 2
disk_present: true
disk {
serial: "Z4D4ATKH"
model: "ST6000NM0024-1HT17Z"
is_mounted: true
current_firmware_version: "SN05"
target_firmware_version: "SN05"
background_operation: ""
vendor_info: "Not Available"
raw_capacity: 6001175126016
}
logical_slot_id: 2
}
slot_list {
location: 3

```

```
disk_present: true
disk {
  serial: "Z4D4AN9J"
  model: "ST6000NM0024-1HT17Z"
  is_mounted: true
  current_firmware_version: "SN05"
  target_firmware_version: "SN05"
  vendor_info: "Not Available"
  raw_capacity: 6001175126016
}
logical_slot_id: 3
}
bios {
  model: "0824"
  firmware_version: "20151207"
}
bmc {
  model: "X10_ATEN"
  firmware_version: "03.63"
}
hba_list {
  index: 1
  model: "LSI Logic SAS3008"
  current_firmware_version: "MPTFW-16.00.10.00-IT"
  target_firmware_version: "MPTFW-16.00.10.00-IT"
}
motherboard {
  model: "X10DRT-P-NI22"
}
hostbootdisk {
  device_model: "SATADOM-SL 3IE3 V2"
  serial_number: "YCA11905240320106"
}
```

## Bad disk handling by Hades

- If Stargate finds that it takes more than 40 seconds to finish a read or write operation to a particular drive, it will be marked offline and given to the Hades service to perform a smartctl health test.



- If the drive passes the health test, it will be marked back online.
- The results of this test will be found in output of "edit-hades -p" on the local CVM.
- If the drive hangs again the same thing will happen unless the drive goes offline 4 times within 1 hour.
- In this case, the drive will be marked permanently offline with the "is\_bad: true" flag even if it passes the smartctl short test.
- In such a case, the disk serial number will be placed in the tombstone\_list, a disk removal task will be initiated, and finally the disk will be unmounted from the CVM.

# Acropolis

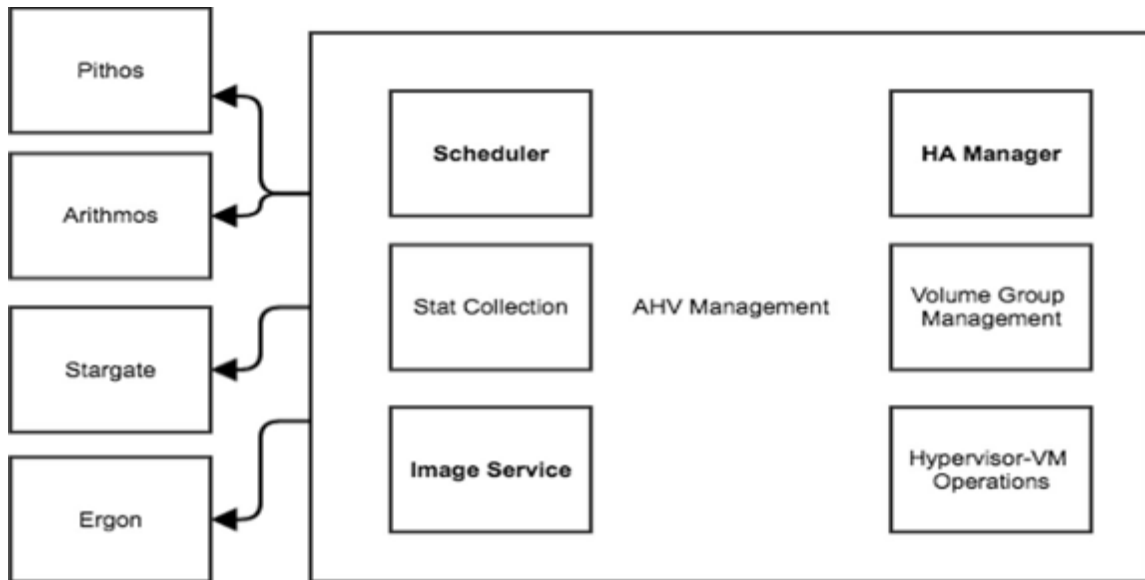
Port: 2030

Acropolis is the service that runs on the every CVM in the cluster with an elected Acropolis Leader which is responsible for task scheduling, execution, IPAM, etc. and it interacts with libvirt on the AHV host for all of the VM management operations. Similar to other components which have a Leader, if the Acropolis Leader fails, a new one will be elected.

## Responsibilities under Acropolis:

- Maintenance mode - Evacuate all the VMs off a host - user triggered (or) upgrade workflows
- Parcel - virtual construct for allocating resources (independent of VM operations)
- HA
  - Reservation types - Best Effort, Reserved Segments
  - Affinity - (must rule, vm to hosts), Anti-Affinity(should rule, vm to vm, within vm\_group)
  - Agent VM - affined to a host
- Lazan (CPU, storage controller hotspots) - Runs periodically (every 15 minutes - lazan\_anomaly\_detection\_period\_secs)

## Acropolis Leader and it's duties



- Acropolis works on Follower ↔ Leader architecture.
- Most of the work is done by the Leader.
- Acropolis master communicates with libvirt in all of the nodes.
- At any point, a host can only be in – connected, Reconciled (or) Disconnected state
- Iptable rules(on host) are managed by acropolis. Should be refreshed
  - Whenever a new master is elected
  - Whenever master reconnects with libvirt
- OVS bridge creation, virtual switch consistency is handled by acropolis
- Processes the batch request sent by other nodes.
- Local nodes are responsible for publishing their stats(host usage stats every 30 seconds) to Arithmos

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## Acropolis Debugging

- Check if acropolis is stable
  - `links -dump http:0:2030 | grep -i master`
  - Look at acropolis master logs - `/home/nutanix/data/logs/acropolis.out`
- Find acropolis master: `service=/appliance/logical/pyleaders/acropolis_master; id=$(zkat $service/ zkls $service| head -1); zeus_config_printer | grep -B 15 " uuid: \\\"$id\\\"" | grep "service_vm_external_ip" | awk '{print "Acropolis leader ==>", $2}'`
- `acli host.list` - See if hosts are schedulable (or) if it's a problem with one host.
- Monitor the leadership changes using : `grep "Acquired master leadership" ~/data/logs/acropolis.out`
- OVS related logs:
- `/var/log/openvswitch` and `/var/log/acropolis_ovs.log`

## Lazan/ADS (Acropolis Dynamic Scheduler) -<Trim down content>

Port: 2038

- Acropolis Dynamic Scheduling (ADS) proactively monitors your cluster for any compute and storage I/O contentions or hotspots over a period of time.

- If ADS detects a problem, ADS creates a migration plan that eliminates hotspots in the cluster by migrating VMs from one host to another.
- Lazan is the ADS service in an AHV cluster
- Lazan is also involved in GPU modelling of GPU based cluster. It participates in power on of VM in GPU based cluster.

**Resource scheduling can be broken down into two key areas:**

- Initial placement - Where an item is scheduled at power-on
- Runtime Optimization - Movement of workloads based upon runtime metrics

## Features of ADS

- ADS is not a load balancer
- ADS also monitors storage controller (stargate) load apart from CPU usage of the cluster
- Does not monitor memory or network.
- No configuration required - Enabled by default

Checking the status of ADS:

```
accli ads.get
nutanix@NTNX-16SM65230111-A-CVM:10.136.106.126:~$ accli ads.ge
t
config {
enabled: True
}
```

To disable ADS: `accli ads.update enable=false`

To enable ADS: `accli ads.update enable=true`

Lazan queries Acropolis for VM-Host mappings, VG details etc. and arithmos for resource usage.

**Logs can be found in:** `~/data/logs/lazan.out`



# Uhura

Port: 2037

VM management back-end service. In case of ESXi as the hypervisor, it is generally used to communicate the cluster's VM operations to the vCenter via APIs.

Along with this, it also manages VM customization such as Sysprep for Windows VMs and Cloudinit for Linux VMs.

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